Paleomagnetic constraints on the tectonic history of the major blocks of China duing the Phanerozoic*

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Abstract Paleomagnetic study of China and its constraints on Asia tectonics has been a hot spot. Some new paleomagnetic data from three major blocks of China, North China Block (NCB), Yangtze Block (YZB) and Tarim Block (TRM) are first reported, and then available published Phanerozoic paleomagnetic poles from these blocks with the goal of placing constraints on the drift history and paleocontinental reconstruction are critically reviewed. It was found that all three major blocks were located at the mid-low latitude in the Southern Hemisphere during the Early Paleozoic. The NCB was probably independent in terms of dynamics, its drift history was dominant by latitudinal placement accompanying rotation in the Early Paleozoic. The YZB was close to Gondwanaland in Cambrian, and separated from Gondwanaland during the Late-Middle Ordovician. The TRM was part of Gondwanaland, and might be close to the YZB and Australia in the Early Paleozoic. Paleomagnetic data show that the TRM was separated from Gondwanaland during the Late-Middle Ordovician, and then drifted northward. The TRM was sutured to Siberia and Kazakstan blocks during the Permian, however, the composite Mongolia-NCB block did not collide with Siberia till Late Jurassic. During Late Permian to Late Triassic, the NCB and YZB were characterized by northern latitudinal placement and rotation on the pivot in the Dabie area. The NCB and YZB collided first in the eastern part where they were located at northern latitude of about 6°—8°, and a triangular oceanic basin remained in the Late Permian. The suturing zone was located at northern latitude of 25° where the two blocks collided at the western part in the Late Triassic. The collision between the two blocks propagated westward after the YZB rotated about 70° relative to the NCB during the Late Permian to Middle Jurassic. Then two blocks were northward drifting (about 5°) together with relative rotating and crust shortening. It was such scissors-like collision procedure that produced intensive compression in the eastern part of suturing zone between the NCB and YZB, in which continental crust subducted into the upper mantle in the Late Permian, and then the ultrahigh-pressure rocks extruded in the Late Triassic. Paleomagnetic data also indicate that three major blocks have been together clockwise rotating about 20° relative to present-day rotation axis since the Late Jurassic. It was proposed that Lahsa Block and India subcontinent successively northward subducted and collided with Eurasia or collision between Pacific/Philippines plates and Eurasia might be responsible for this clockwise rotating of Chinese continent.

Keywords: China, paleomagnetic, tectonophysics.

China continent, which covers 9.6 million km^2 in eastern Asia, consists of the North China Block (NCB), the Yangtze Block (YZB), the Tarim Block (TRM) and the Lahsa Block and so on. These blocks are separated by Paleozoic and Mesozoic accretionary belts and/or large faults (fig. 1), and comprised a tectonically complicated area[1,2]. There are complex aspects of

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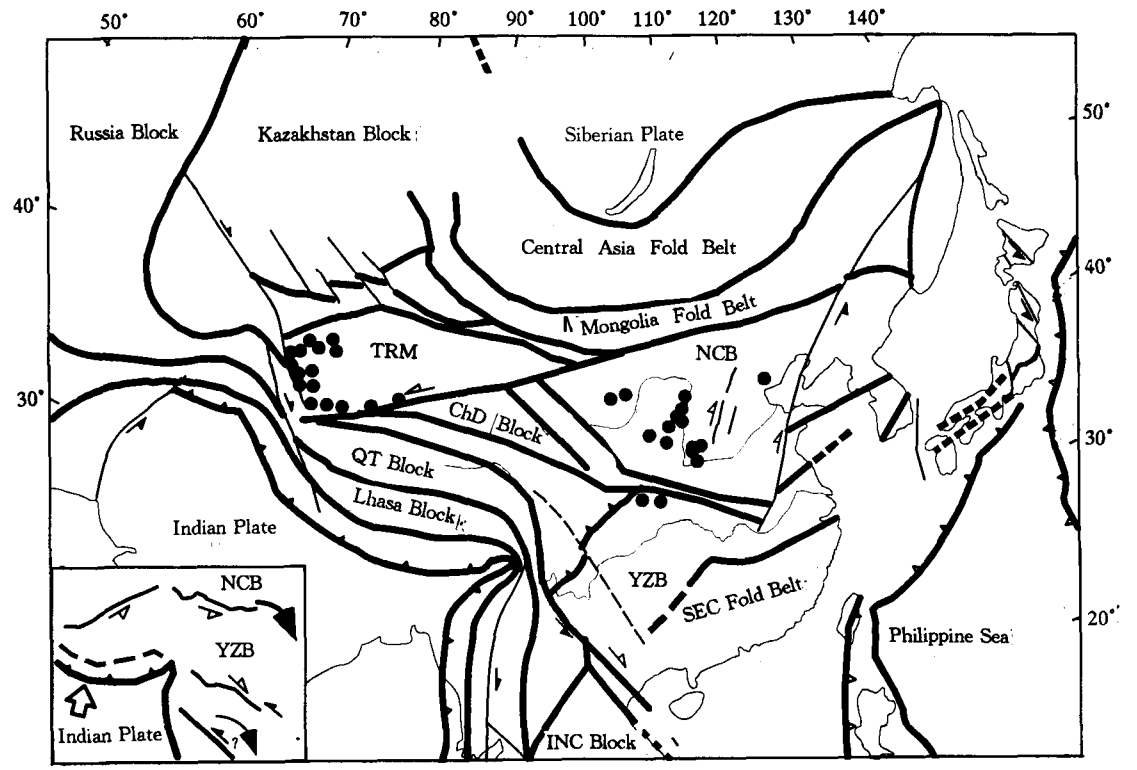


Fig. 1. Schematic block map of China showing paleomagnetic sampling sites by solid circles. ChD, Chaidamu; QL, Qilian; QT, Qiangtang; S-P, Song-Pan; SEC, Southeast China.

intercontinental collision and intracontinental deformation. It has been revealed that the basement age of the NCB and YZB are quite different while that of the YZB and TRM is similar[3,4]. Although it has been suggested that the YZB was separated from Gondwanaland during the Late Precambrian[5], a close biogeographic association of the YZB with Australia has recognized for both the Cambrian and Devonian[6,7], indicating that the YZB may have been part of Gondwanaland in Early Paleozoic. In addition, Cambrian and Ordovician Bradoriida fossils from both the TRM and YZB are very similar, suggesting that the TRM was also part of Gondwanaland during the Early Paleozoic[8,9]. Whether the NCB also belonged to Gondwanaland during this period[10] or not remains debatable.

Paleomagnetic data have provided some evidence concerning the times and modes of collisions of major China's blocks during the Late Paleozoic and Mesozoic[11—26], showing obvious relative motions between the NCB, YZB and TRM. But China's tectonic history and manner of amalgamation and suturing is still disputed. To understand the Chinese tectonic history a multidisciplinary approach (such as paleontology, radiochronology, petrology, structural geology, etc.) is required. In this paper, we mainly present paleomagnetic constraints on the geodynamic history of China during the Phanerozoic. We review and critically assess all the available paleomagnetic data published since the beginning of the 1980s. Delimiting its spatial and temporal applicability, we have mainly made our effort in filling the paleomagnetic data in the blank spots in the Early Paleozoic for the NCB, YZB and TRM. New paleomagnetic data are throwing considerable light on refinement of the APWPs of the three major blocks, also helpful to clarifying the kinematics

and dynamics history of these blocks.

1 Regional geology and paleomagnetic sampling

1.1 North China Block

The Cambrian to Middle Ordovician strata of the NCB were well developed and belonged to platform sediments, which consisted mainly of marine clasolite and carbonate. Due to the regional crust uplifting in Late Ordovician, there was a sedimentary hiatus from Late Ordovician through Early Carboniferous in the NCB. Almost complete strata have been developed in the NCB since Middle Carboniferous. Limestone was sampled from the Middle Ordovician Fengfeng, Majiagou and Lower Ordovician Liangjiashan Formation at Jiaozuo (35.3°N/113.2°E) of Henan Province[27]. Paleomagnetic samples from the Middle Ordovician Maozhuang Formation red clasolite, Lower Cambrian Xinji Formation, Middle Cambrian Zhangxia Formation, Upper Cambrian Fengshan Formation and Middle Ordovician Majiagou Formation were collected at Jiyuan (35.2°N/112.7°E) of Henan Province and Jincheng (35.3°N/113.2°E) of Shanxi Province[27,28]. Also, Lower Cambrian Xinji Formation in Henan Province, Middle Cambrian Zhangxia Formation, Upper Cambrian Fengshan Formation and Middle Ordovician Majiagou Formation were sampled at Dengfeng of Henan Province (34.5°N/113°E). A total of 979 oriented core samples and 28 hand samples from 98 sites were collected using a portable drill. In order to compare these results, 917 oriented core samples from 124 sites were collected using a portable drill at Helanshan areas in Ningxia, Hancheng and Tongchuan in Shaanxi Province, and Luliang-Yuncheng region in Shanxi Province, which covers from the Early Cambrian to Middle Ordovician. The lithological character includes limestone, dolomitic limestone, pelitic limestone, pelitic strip limestone, dolomite, domicrite, pelitic dolomite and quartzose sandstone[29].

1.2 Yangtze Block

The Xingshan-Zigui section of Hubei Province was selected as the sampling region, where strata sequence is relatively complete and clear. We collected oriented core samples from well-developed natural outcrop using a portable drill, covering the Early Cambrian to Upper Jurassic. Strata of Early Cambrian to Early Triassic consist of marine deposits including argillite-carbonate rocks in Cambrian, carbonate and argillite-clasolite rocks between Ordovician and Devonian, carbonate rock between Permian and Early Triassic. All these marine sediments were formed in shelf sea-open platform. Strata of Middle Triassic and Late Triassic consist of continental deposit bore coal. Variegated clasolite is dominant in the Jurassic, and red clasolite between the Cretaceous and Tertiary. These continental sediments were formed in the environment of large inland basin transforming to river and limnetic environment. Two sections (one is from Xinshan to Zigui road and the other is along the Baisha River around Xinshan County (110.38°E/31.15°N) of Sichuan Province) were sampled. Another section was arranged at Daxiakou about 20 km away from Xinshan County. A total of 1000 oriented core samples from 84 sites were collected using a portable drill, which covers Early Cambrian to Late Jurassic. In order to compare these results, Wangcang-Nanjiang section (106.4°E/32.1°N) was sampled along a road, where Phanerozoic strata were complete and formed a short axis anticline. The main lithological characters are sandstone, mudstone and limestone. 1190 oriented core samples from 107 sites were collected using a portable drill.

1.3 Tarim Block

The TRM is located in Xinjiang Uygur Autonomous Region, extending the EW direction as a rhomb in shape. The Tarim Basin is composed of the main part of the TRM. The Ordovician and Silurian occur along the northern periphery of the basin. The Silurian, Devonian, Carboniferous and Permian are presented at Keping areas in the northwestern part of the basin. The lithological sorts are composed mainly of dolomitic limestone, red and yellow siltstones. The Triassic and Jurassic, consisting mainly of green sandstone and red mudstone, are distributed in the west of the Kuche depression. The Mesozoic and Cenozoic rocks are presented with the same lithology in the middle part of the depression[30]. The strata from the Devonian to the Quaternary are distributed in southwestern and southeastern depressions and Tie-ke-li-ke uplift. The Cretaceous and Cenozoic consist mainly of volcanic rocks, and the others are composed of limestone, sandstone and mudstone[30]. About 980 oriented core samples from 164 sites were collected using a portable drill.

2 Experiment and results

The core samples, 2.5 cm in diameter, were cut into 2.2-cm-long specimens for paleomagnetic measurements. Also, small chips were cut for rock magnetic studies. In order to identify the magnetic minerals, its grain size and concentration, multi-rock magnetic studies were carried out, such as acquisition of isothermal remanent magnetization (IRM), remanent coercivity, magnetic hysteresis loops, thermomagnetic analysis, thermal demagnetization of three-components IRM.

All specimens were subjected to progressive thermal/alternating field demagnetization. Demagnetization and measurement at each stage of treatment by 2G cryogenic magnetometer at the Paleomagnetic Laboratory of the Institute of Geophysics of CAS, the Institute de Physique du Globe de Paris, the University of California at Santa Cruz. The Bulk susceptibility of the specimens was measured after each heating step, in order to monitor possible magnetic mineral transformations. Magnetic directions were plotted on Zijderveld vector diagrams to identify discrete magnetic components by principal component analysis[31]. Mean directions were calculated using Fisher statistics when directions were only obtained within a site, or by McFadden and McElhinny's method when individual characteristic directions had to be combined with great circle intersections with sector constraints. A single component characteristic remanent magnetization (ChRM) was isolated at higher temperature for some sampling sites. Most of data pass fold and/or reversal test. Some new paleomagnetic data were obtained from the NCB, YZB and TRM (see table 1).

3 APWPs for NCB, YZB and TRM and their tectonic implication

3.1 APWPs of the NCB, YZB and TRM during the Phanerozoic

Table 1 shows our new paleomagnetic data obtained from the three major blocks, and detailed analysis of the data will be presented in the following papers. Also, we have critically reexamined the available paleomagnetic database, 116, 90 and 34 paleopoles have been collected respectively from the NCB, YZB and TRM during the Phanerozoic, which satisfied reliability tests[32—34]. Statistic results are shown in table 2.

The synthetic paleomagnetic results in table 2 indicate that (i) Cenozoic poles for both NCB and YZB are statistically consistent with present geographic pole, showing a similar configuration

Table 1 New paleomagnetic data obtained from NCB, YZB and TRM

Block	Age	ϕ_s/(°E)	λ_s/(°N)	N	n	D_s/(°)	I_s/(°)	k	α_{95}/(°)	ϕ_p/(°E)	λ_p/(°N)	dp/(°)	dm/(°)	p/(°N)	Comment
	Q	109.2	34.2	12	356	−175.7	−55.6	452	1.9	169.1	86.0	1.9	2.7	36.1	
	O_2	113.3	35.3	6	49	163.4	42.4	39.0	9.2	310.4	28.1	7.0	11.4	−24.5	F
	O_2	109.0	36.1	5	29	329.8	−26.7	41.6	12	32.2	324.2	7.1	13	−14.1	
	O_1	110.6	35.6	9	41	333.6	−20.2	32.6	9.2	324.1	37.6	5.1	9.6	−10.4	F
NCB	Cam/O_1	117.9	39.6	1	49	182.9	32.2	19.6	4.7	294.6	32.9	2.9	5.1	−17.5	F
	Cam_3	110.5	35.6	10	56	326.9	−25.8	73.2	5.7	329.1	31.7	3.3	6.2	−13.6	F
	Cam_2	123.4	35.7	2	20	159.0	62.7			318.5	8.0				F
	Cam_2	110.5	35.6	11	59	324	−19.4	88	4.9	334.2	33.2	2.7	5.1	−10.8	F
	Cam_1	110.5	35.6	8	34	308.7	−32.3	64.1	7.0	342.8	18.8	4.5	7.9	−17.5	F
	J_3	110.6	31.3	4	49	33.1	41.1	86.8	9.9	206.7	59.8	7.3	12.1	23.6	F
	J_2	110.6	31.3	7	91	356.5	44.3	23.4	12.7	322.1	84.1	10.1	15.9	26.1	F
	J_1	110.6	31.3	9	140	351.7	39.6	107.8	5.1	332.8	78.6	3.7	6.1	22.5	F
YZB	T_2	110.6	31.3	7	91	203.8	−45.3	17.7	14.7	206.2	206.2	11.8	18.0	26.8	F
	T_1	110.6	31.3	3	41	227.6	−16.8	23.1	26.3	216.3	39.3	2.4	4.6	8.4	F
	P_2	110.6	31.3	2	15	194.7	−22.3	85.5	4.1	253.1	67.1	2.4	4.0	12.1	F
	S_1/S_2	110.6	31.3	6	41	215.9	11.4	64.5	8.4	61.1	−39.6	4.3	8.5	−5.8	F
	$O_{1\text{-}2}$	110.6	31.3	4	35	146.2	17.6	45.7	13.7	154.9	−38.4	7.4	14.2	−9.1	R
	Cam	106.3	32.2	9	74	129.1	−23.6	44.6	7.8	185.1	−39.5	4.4	8.3	−12.3	F
	Q	81.5	36.2	3	120	191.9	−53.0	4968	1.7	183.1	79.9	1.6	2.4	33.6	
	E_1	81.5	36.2	4	24	65.8	63.2	92	10.1	137.1	42.6	11.4	14.5	44.7	F
	K_2	82.9	42	10		15.0	35.0	19	11	229.0	64.0	7.3	12.7	19.3	R
	K_1	78.8	36.3	8	52	195.3	−42.3	12.9	16.8	211.6	69.7	9.8	15.9	24.5	R
	J_2	79.2	36.0	10	53	216.7	−35.3	56	6.6	187.9	52.4	3.7	6.5	19.5	R
TRM	J_2	83	41.8	7		228.8	−49.3	62.0	8.3	173.0	49.9	2.8	4.1	30.2	F
	T_3	80.5	41.8	8		227.5	−53.2	87.4	5.8	166.8	52.1	5.6	8.0	33.8	R
	T_2	81.5	40.9	1	26	226.5	−52.5	42.2	4.4	168.2	52.5	4.2	6.1	33.1	R
	T_1	74.9	41	5		221.4	−44.3	103.1	8.0	175.5	52.8	6.3	10.1	26.1	R
	S	79.7	40.5	6		87.9	30.4	74.6	8.1	158.4	12.1	4.1	7.2	16.2	B
	O_1	83.4	41.3	5		101.2	−33.9	38.2	13.1	180.6	−20.4	8.5	15.0	−18.6	C

ϕ_s and λ_s, the position of sampling site; N, number of sampling sites; n, number of samples; D_s/I_s, declination/inclination in the stratigraphical coordinates; k, precision parameter after tilt correction; α_{95}, half-angle of cone of 95% confidence for paleomagnetic directions; ϕp and λp indicate the position of pole; dp and dm, oval of 95% confidence for the paleomagnetic pole; p, paleolatitute; F, fold test; R, reversal test; B, baked test; C, uniform test.

Table 2 Average paleopoles from NCB, YZB and TRM

Block	T/(Ma)	N	ϕp/(°E)	λp/(°N)	$A_{95}(\alpha_{95})$/(°)	D/(°) ± dD	P/(°) ± dP	DD/(°)	DP/(°)	DD/dt	DP/dt
NCB	1	12	169.1	86.0	2.7(2.8)	5.0±4.46	41.2±2.55				
	32	5	202.9	88.7	4.7(2.6)	1.6±3.88	39.7±2.22	-3.4	1.5	-0.11	0.048
	105	3	204.9	80.7	9.5(6.1)	11.1±8.22	36.8±4.74	-9.5	2.9	0.131	0.04
	151	2	221.8	69.4	15.4(7.9)	19.1±7.88	26.9±4.46	8	9.9	0.171	0.215
	172	5	229.7	76.8	5.6(3.8)	11.4±4.53	30.6±2.6	-7.7	-3.7	-0.37	-0.176
	193	1	286.0	82.4	6.8(3.9)	-1.3±4.41	32.5±2.54	-12.7	-1.9	-0.6	-0.09
	221	1	7.7	62.3	3.8(4.8)	-34.4±5.72	34.6±3.3	-33.1	-2.1	-1.18	-0.075
	238	7	2.0	61.0	2.9(3.5)	-34.5±4.33	31.6±2.49	-0.01	3	-0.01	0.176
	243	6	353.7	56.9	4.6(3.6)	-36±3.9	25.8±2.22	-1.5	5.8	-0.3	1.16
	250	11	355.1	50.3	5.7(4.3)	-42.4±4.43	22.7±2.46	-6.4	3.1	-0.91	0.443
	481	4	314.2	32.9	9.6(8.9)	-30.2±7.15	-9.8±3.67	12.2	32.5	0.05	0.141
	513	16	323.0	28.3	7.3(6.8)	-39.2±5.69	-9.9±2.93	-9	0.1	-0.28	0.003
	526	26	324.1	30.6	7.4(4.0)	-38.7±6.04	-7.5±3.07	0.5	-2.4	0.04	-0.185
	553	17	332.4	20.6	10.3(5.2)	-50.8±7.21	-11.0±3.73	-12.1	3.5	-0.45	0.13
YZB	1	17	328.7	89.4	5.9(3.1)	359.4±4.74	39.6±2.72				
	105	14	209.0	79.4	4.6(2.9)	12.2±3.99	35.5±2.31	12.8	4.1	0.12	0.039
	151	6	201.6	71.3	8.6(4.7)	21.9±6.08	33.4±3.51	9.7	2.1	0.22	0.046
	172	7	204.0	72.0	9.0(4.5)	20.8±5.77	33.1±3.32	-1.1	0.3	-0.05	0.014
	193	1	332.8	78.6	5.1(2.6)	348.9±2.98	32.9±1.72	-31.9	0.2	-1.52	0.01
	221	5	187.2	51.4	8.5(4.9)	46.5±3.11	30.7±1.78	57.5	2.2	2.05	0.079
	238	3	214.7	56.3	20.3(10.8)	31.9±10.0	20.1±5.48	-14.5	10.6	-0.85	0.624
	243	11	216.7	44.5	4.7(2.4)	39.6±2.03	10.8±1.05	7.8	9.3	1.56	1.86
	245	8	220.0	47.7	5.8(3.8)	35.6±3.21	11.7±1.66	-4	-0.9	-2	-0.45
	250	13	243.6	55.3	5.2(5.1)	19±4.84	9.7±2.49	-16.6	2	-3.32	0.4
	273	1	265.2	65.3	8.1(6.8)	5.5±5.47	15.8±2.91	-13.5	-6.1	-0.59	-0.265
	416	1	195.7	6.8	5.3(2.7)	79.8±3.43	-1.5±1.72	74.3	17.3	0.52	0.121
	431	1	157.0	-55.7	8.0(5.3)	149.5±4.42	-18±2.39	69.7	16.5	4.65	1.1
	481	1	154.9	-38.4	14.2(8.7)	138.8±6.15	-4.1±3.09	-10.7	-13.9	-0.214	-0.278
	526	1	185.1	-39.5	8.3(8.1)	123.6±7.33	-22.3±4.06	-15.2	18.2	-0.34	0.404
TRM	1	1	183.1	79.9	2.4(1.7)	13.2±2.33	40.1±1.34				
	32	2	192.2	73.4	8.9(6.2)	20.9±7.82	36.9±4.51	7.7	3.22	0.25	0.103
	105	1	211.6	69.7	15.9(8.0)	21.5±8.5	29.8±4.87	0.6	7.1	0.01	0.097
	172	3	194.2	74.3	23.0(13.0)	19.6±17.48	36.6±10.09	-1.9	-6.8	-0.03	-0.101
	238	4	182.3	66.2	13.3(12.1)	30.7±17.16	38.2±9.88	11.1	-1.6	0.17	-0.024
	250	9	183.6	60.0	4.5(2.3)	38±5.41	35.9±3.12	7.3	2.3	0.61	0.192
	306	4	175.7	48.7	3.7(1.9)	53.1±4.18	36.3±2.41	15.1	-0.4	0.27	-0.007
	390	2	153.7	10.0	1.9(1.1)	105.4±2.15	32.5±1.24	52.3	3.8	0.62	0.045
	427	3	162.9	13.1	5.6(4.0)	96±4.33	27.5±2.46	-9.4	5	-0.25	0.135
	481	3	181.5	-20.1	33.8(17.1)	109.6±13.32	-8.0±6.79	13.6	35.5	0.25	0.657

N, Number of pole; T, time; A_{95}, half-angle of cone of 95% confidence of poles; D/P, declination/paleolatitude; dD/dP, 95% confidence of declination/paleolatitude; DD/DP, amounts of rotation and placement; $\frac{DD}{dt}$/((°)·Ma^{-1}), amount of rotation for a million year; $\frac{DP}{dt}$/((°)·Ma^{-1}), amount of placement for a million year, reference point is placed at 40°N/98°E.

with the present between these two blocks in this period. Tertiary paleomagnetic pole of the TRM is significantly discrepant with geographic pole. Paleolatitude of the TRM in the Tertiary is obviously lower than that deduced from Eurasian reference poles, which might result from intercontinent shortening. (ii) Except for the Cretaceous, the resolution for Mesozoic paleomagnetic data are at the stage level (e.g. lower, middle and upper stage of geologic period) with better reliability for both the YZB and NCB than the TRM, but only sub-era level for the TRM. (iii) There still are no reliable paleomagnetic poles of the Silurian, Devonian and Carboniferous for the NCB, Devonian and Carboniferous poles for the YZB, and Cambrian pole for the TRM.

New APWPs of the major blocks (fig. 2) obtained using China's poles in table 2 give the following conclusions: (i) New APWPs are generally consistent with the published APWP for the NCB and YZB from the Late Permian to Quaternary[11—26], but significant difference exists in pre-Permian for the two blocks. (ii) Reliability of paleomagnetic data obtained from the TRM is not better than that of the NCB and YZB. (iii) APWPs of the NCB, YZB and TRM are obviously discrepant from the Paleozoic to Triassic, which would not be eliminated by simple rotating,

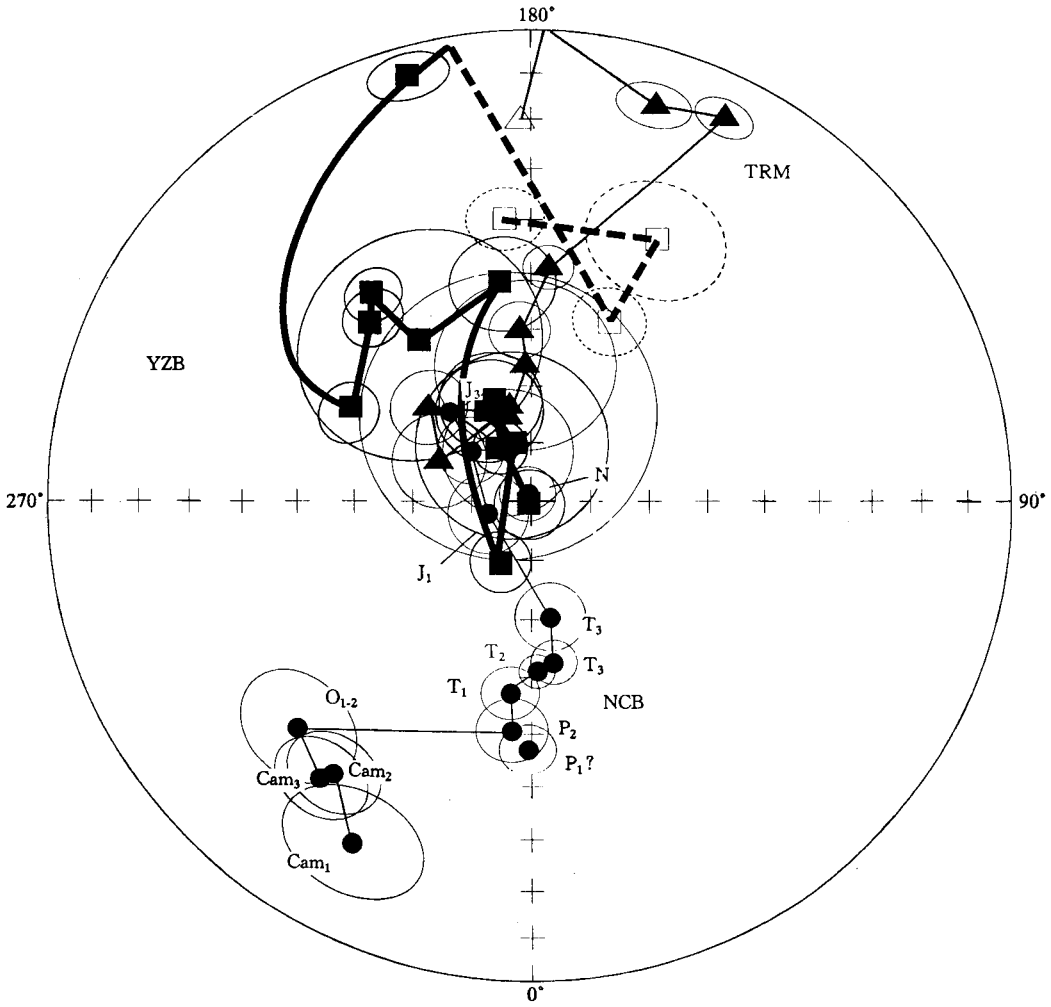


Fig. 2. APWPs for the three major blocks.

demonstrating that they are dynamically separated from each other in this period. The amalgamations between them could not be complete (details discussed later). APWPs for the three blocks have become consistent since the Late Jurassic, showing that they have completely sutured and no obvious relative displacement exists between them. It suggests that the three blocks may have been dynamically a composite block. It should be emphasized that kinematics and dynamics history revealed by APWPs are reflected by both the systematic drift history of blocks before collision and crust shortening and relative rotation after suturing. Thus APWPs would suggest when the different blocks completely become a dynamically united block. It does not seem to reveal when the different blocks amalgamated if we have not other geologic evidence.

3.2 Drift history for the three major blocks

The drift history of a block can be inferred in terms of its paleodeclination (D) and paleolatitude (P) in different periods. In this paper we have used the following methods to describe the drift history of the three major blocks. First, a reference point at 40°N/98°E of the NCB, YZB and TRM was chosen. Second, the paleodeclination and paleolatitude with their standard deviation at this point were calculated from the paleopoles of the NCB, YZB and TRM (table 2). The orientations of the three blocks were initially determined with respect to the present geographic pole. The amount of rotation (DD) for the two successive periods was calculated by subtracting a younger D value from an older D value, so that a positive value of DD corresponds to a clockwise rotation, and a negative one to an anticlockwise rotation. The amount of latitudinal drift (DP) was determined by subtracting an older P from a younger P value. Positive values of DP indicate northward motion and negative ones southward motion. The drift data and drift history of the NCB, YZB and TRM in successive periods are presented in table 2. The ages are following the time scale of Harland et al.[35].

Based on the assumption of a geocentric dipole field, the drift and tectonic relationships between two blocks can be inferred by comparison of their corresponding rotation and latitudinal displacement. To clarify when and in what aspects the NCB and YZB scissors-like collided and sutured together, we have calculated paleodeclination (relative rotation between two blocks) and paleolatitude (relative latitudinal displacement) using two reference points which are located at Yuexi area (31°N/116.5°E) and the Fengzhou area (34°N/106.5°E), respectively. The results calculated above are shown in tables 3 and 4. The results from table 3 are further demonstrated in fig. 3. It can be seen that the NCB (reference point) was located at the mid-low southern latitude from the Cambrian to Ordovician (fig. 3(a), (b)), and the declinations corresponding to the reference point are northwestward (fig. 3(c), (d)). The NCB across the equator and located at low northern latitude in the Late Paleozoic indicated that it had northerly drifted 3000 km from the Early Cambrian to Late Permian but the declinations maintained northwestward in these periods (fig. 3(c)). Latitudinal displacement of the NCB, with basically constant drift rate, was northerly in the Triassic, and the declination was also unchanged. All these behavior might imply that the NCB was a dynamically isolated block from the Cambrian to Late Triassic, which is supported by geologic and paleontologic studies[36]. However, a rapid anticlockwise rotation for the NCB occurred from the Late Triassic to Middile Jurassic (fig. 3(c), (d)). Therefore, it is obvious that the latitudinal displacement was dominant in the pre-Late Triassic period and then rotation occurred for the NCB.

Table 3 Paleodeclination and paleolatitude deduced from the paleopoles of the NCB and YZB at reference points

Period	NCB				Period	YZB			
	east reference point Yuexi (31°N/116.5°E)		west reference point Fengzhou (34°N/106.5°E)			east reference point Yuexi (31°N/116.5°E)		west reference point Fengzhou (34°N/106.5°E)	
	P/(°N)	D/(°)	P/(°N)	D/(°)		P/(°N)	D/(°)	P/(°N)	D/(°)
Q	33.4	3.8	35.8	4.4					
N	31.1	1.5	33.8	1.6	Q-Ter	30.5	-0.4	33.6	-0.5
K	30.8	10.8	32.2	10.9	K	30	12.2	31.1	12.1
J_3	23.7	21.8	23.5	20.3	J_3	30.8	21.8	30.4	21.7
J_2	25.1	13.4	26.2	12.3	J_2	30.1	20.9	29.9	20.7
J_1	23.5	1.5	26.4	0.1	J_1	21.6	-7.2	25.8	-9.1
T_3	19.1	-27.8	25.9	-30.5					
T_2	16.1	-27.3	22.9	-30.6	T_2	21.1	36.1	18.8	33.8
T_1	10.2	-27.8	17	-31.8	T_1	14.6	46.5	10.8	43
					P/T	14.3	42.5	11	39
P_2	6.4	-33.3	13.7	-37.7	P_2	7.9	27.3	6.5	23
					P_1	9.3	12.7	10.7	8.9
					S_{2-3}	12.7	89.7	4.5	84.8
					S_{1-2}	-3.3	158.5	-9.5	153.8
O_{1-2}	-23.9	-16.2	-18.2	-24.3	O	11.9	150.2	4.8	144
Cam_3	-25.5	-25.8	-18.8	-33.6					
Cam_2	-23.1	-25.7	-16.3	-33.2	Cam	-4.9	133.9	-13.3	129
Cam_1	-28	-38.4	-20	-45.7					

Table 4 Paleodeclination and paleolatitude deduced from the paleopoles of the NCB, YZB and TRM at reference points

Period	Reference point (40°N/98°E)				Period	Period	Reference point (31°N/104°E)				Period
	NCB		TRM				YZB		TRM		
	P/(°N)	D/(°)	P/(°N)	D/(°)			P/(°N)	D/(°)	P/(°N)	D/(°)	
Q	41.2	5.0	40.1	13.2	Q	Q	30.6	-0.5	32.4	11.8	Q
N	39.7	1.6	36.9	20.9	N			30.1	19.3	N	
K	36.8	11.1	29.8	21.5	K	K	27.7	11.6	23.1	21.1	K
J_3	26.9	19.1				J_3	20.8	20.9			
J_2	30.6	11.4	36.6	19.6	J	J_2	26.3	19.9	29.7	18.1	J
J_1	32.5	-1.3				J_1	23.2	-9.3			
T_3	34.6	-34.4				T_3	27.8	44.4			
T_2	31.6	-34.5	38.2	30.7	T	T_2	15.1	32.5	32.8	28	T
T_1	25.8	-36				T_1	7.2	41.5			
P_2	22.7	-42.4	35.9	38	P	P/T	7.4	37.6			
			36.3	53.1	C_3	P_2	3	21.6	31.6	35.3	P
			32.5	105.4	D_{1-2}				34.4	49.4	C_3
			27.5	96.0	S				39.5	103.4	D_{1-2}
O_{1-2}	-9.8	-30.2	-8	109.6	O_1	S_{2-3}	2	83.3			
Cam_3	-9.9	-39.2				S_{1-2}	-7.7	153	33.2	94.4	S
Cam_2	-7.5	-38.7				O	6	142.3	-0.2	113.5	O_1
Cam_1	-11	-50.8				Cam	-13	128.8			

The drift and displacement history of the YZB is also shown in fig. 3. The YZB wandered near the equator in the Early Paleozoic and declination maintained SSE, which was obviously different from that of today. Both northern latitudinal displacement (about 1 600 km) and anticlockwise rotation occurred for the YZB from the Cambrian to Ordovician. The southern latitudinal displacement (about 1 500 km) was dominant, but no obvious rotation was observed from the

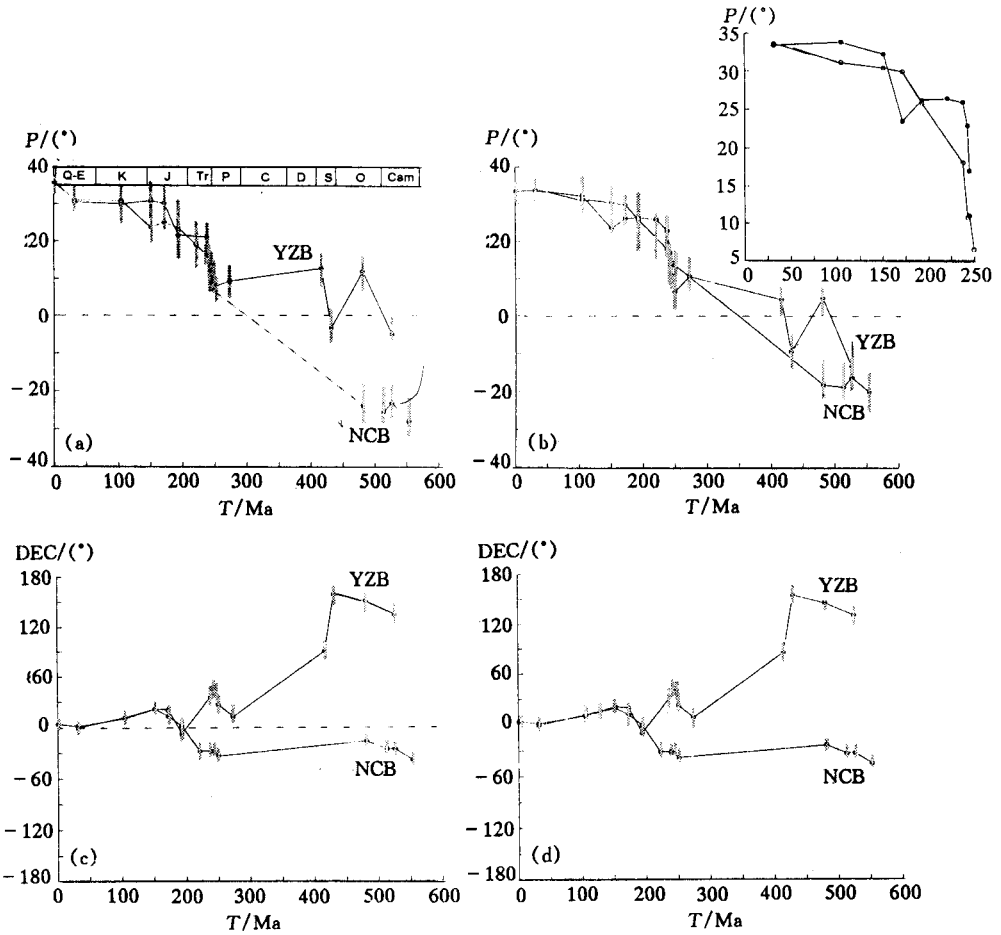


Fig. 3. Paleolatitudes and declinations determined at reference points at Yuexi (31°N/116.5°E) (a, c) and Fengzhou (34°N/106.5°E) (b, d) for the NCB and YZB.

Ordovician to Early Silurian. Relatively rapid northern latitudinal displacement (1 600 km) occurred from the Early to Late Silurian, with a clockwise rotation. Basically, the YZB stayed at low northern latitude, however, with a large clockwise rotation of 77° from the Late Silurian to Early Permian. The drift and displacement history of the YZB in the Early Paleozoic was completely different from that of the NCB, indicating that the two blocks were separated from each other and were not sutured together. It is noted that the ophiolitic mélange and accompanying island-arc type volcanic belt were observed in the Danfeng Group[37], which indicates that some collisions occurred between the mini-blocks in Qinling paleoocean and the NCB during the Caledonian orogeny. Geodynamic action between the NCB and YZB is not excluded in this period. However, this deduction is very tentative for lack of reliable Devonian and Carboniferous paleomagnetic data for both the NCB and YZB.

The latitudinal displacement history and paleolatitude for both the NCB and YZB were very similar in Permian. The drift history of the YZB was characterized by a clockwise rotation, which is significantly greater than that of the NCB from Late Triassic to Early Jurassic, the latter behaved as an anticlockwise rotation from the Early Jurassic to Middle Jurassic.

The paleolatitudes calculated from both the NCB and YZB in the Late Permian, at the reference point at Yuexi (31°N/116.5°E), were consistent at the 95% confidence level (fig. 3(a), (c)), but with a significant difference in declination (60°). The paleolatitudes deduced from both the NCB and YZB at the reference point at Fengzhou of west Qinling (34°N/106.5°E), were different at the 95% confidence level until the Late Triassic (fig. 3(b), (d)). A systematically latitudinal discrepancy (about 4°—7°) exists between the eastern and western reference points located in the Qinling orogenic belt from the Late Permian to Late Triassic (fig. 3). This observation indicates that the two blocks collided first in the eastern part of the central orogenic belt in the Late Permian while the western part occurred in the Late Triassic (fig. 4). It is worth noting that the collision between two blocks does not mean the disappearance of a paleo-ocean, the existence of littoral facies cannot be ruled out. This statement is also supported by the fact that the recession of the middle-low Yangtze River areas occurred in the Middle Triassic, and the Dabie-Sulu ultra-high-pressure metamorphic belt formed in the Late Triassic[38—40]. Paleomagnetic data revealed that the tectonic history of the NCB and YZB was characterized by northern latitudinal displacement and rotation pivoted in the east Qinling from the Late Permian to Late Triassic (fig. 4).

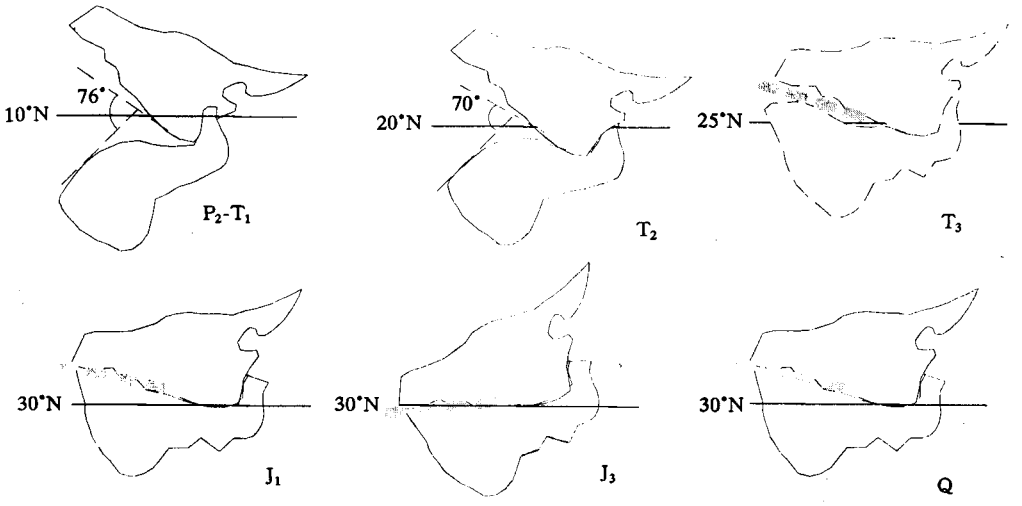


Fig. 4. Collision and suturation procedure of the NCB and YZB.

The westward chap between two blocks was about 70° when they initially collided in the eastern part, while the NCB and Mongolia Block had collided together[10]. The NCB rotated anticlockwise due to a continuous push of the Mongolia Block from the Late Permian to Middle Jurassic. The YZB rotated clockwise due to an inertia forcing driven by itself northward drifting in the same period. After that the NCB and YZB were completely sutured and slowly rotated anticlockwise together. The rotation was probably caused due to the momentum of the NCB and Mongolia Block being much bigger than that of the YZB[41]. This is supported by the fact that huge ultra-high-pressure metamorphic belts at Yuexi in Dabie Mt. was formed in 244—211Ma[39], while no such metamorphic belts were developed in western Qinglin but the formation of huge volcanic magma was of active continental margin type[42]. The Hetaoshan metamorphic volcanic rock, characterized by MORB trace element using the Nd isotopic method, was recently discovered at Mianlüe orogeny in southern Qinling. It was believed that this rock was originally a Paleozoic O-

ceanic fragment, which was metamorphosed and uplifted in the Triassic. The close of Mianlüe Paleo-ocean in the Late Triassic resulted in the suturation between the YZB and NCB in the west part of the central orogenic belt[43].

Angular unconformity between the Middle and Late Triassic was revealed on both sides of Qinling orogeny. Late Triassic strata are absent in the middle Qinglin orogenic belt while the marine deposits appeared in the west Qinling belt. All the observations supported the NCB and YZB collided first in the eastern part of Qinling, and a triangular oceanic basin remained in the Late Permian between the NCB and YZB. The collision propagated westward when the two blocks approached each other with a large rotation of about 70° during the Late Permian and Middle Jurassic. It was such a collision and suturation procedure that produced intensive high compression around the east suturing pivot between the NCB and YZB, continental crust of the YZB subducted into the upper mantle in the Late Permian-Triassic, then the ultrahigh-pressure rocks cropped out during the further penetration between the NCB and YZB. For the collision and suturation procedure of the NCB and YZB and the forming history of the Dabie-Sulu UHP belt, we propose the following tectonic mode on the basis of paleomagnetic data: (i) The collision occurred first in the east, and then west (P_2 in the east, T_3 in the west); (ii) rotated pivot was located in the Dabie Mt. and then moved northward together (from the Late Permian to Late Triassic); (iii) suturation accompanying intensive crust shortening and intracontinent relative rotation during the Late Triassic and Middle Jurassic; (iv) tectonic adjustment with intracontinent relative rotation during the Middle and Late Jurassic.

The tectonic history of the NCB and TRM is of very interest. In the middle of the 1980s paleomagnetic study revealed that for paleolatitudes and declinations determined from the NCB and TRM there was obvious discrepancy in the Permian/Triassic, indicating that the two blocks were separated from each other in this period[44]. The TRM drifted relatively rapidly northward at middle-low northern latitudes (about 3 500 km) from the Early Ordovician to Silurian, and was almost stationary during the Silurian and Early Devonian. The movement was dominant by a clockwise rotation during the Early Devonian to Late Permian (about 67°). Due to a strata hiatus of Silurian-Carboniferous, we cannot describe in detail the drift history for the NCB in this period. However, according to the interpretation of the paleomagnetic data from the Ordovician to Permian, the NCB was located near the equator during the Silurian-Devonian (fig. 5(a)). Therefore, as shown in fig. 5, for the paleolatitudes and declinations determined from the NCB and TRM there was obvious discrepancy during the Silurian-Devonian (fig. 5(b)). The possible ophiolite between the NCB and TRM was dated in Carboniferous-Permian, indicating that they were sutured after that time. Our new paleomagnetic data indicate that the declination and paleopoles determined from the NCB and YZB are consistent at 95% confidence level in the Jurassic, indicating a Jurassic suturation between them.

As shown in fig. 5, the paleolatitudes and declinations calculated from the paleopoles of the TRM and YZB are statistically consistent in the Early Paleozoic, which might indicate that they were in a close position in this period. Geologic results indicated a similar Sinian sediment facies and tillite on both the YZB and TRM. The biota history and sedimentary environments were also similar for the two blocks during the Cambrian and Ordovician. All the geologic evidence implied that the TRM and YZB were in a similar paleogeographic environment and a close position during the Sinian to Ordovician.

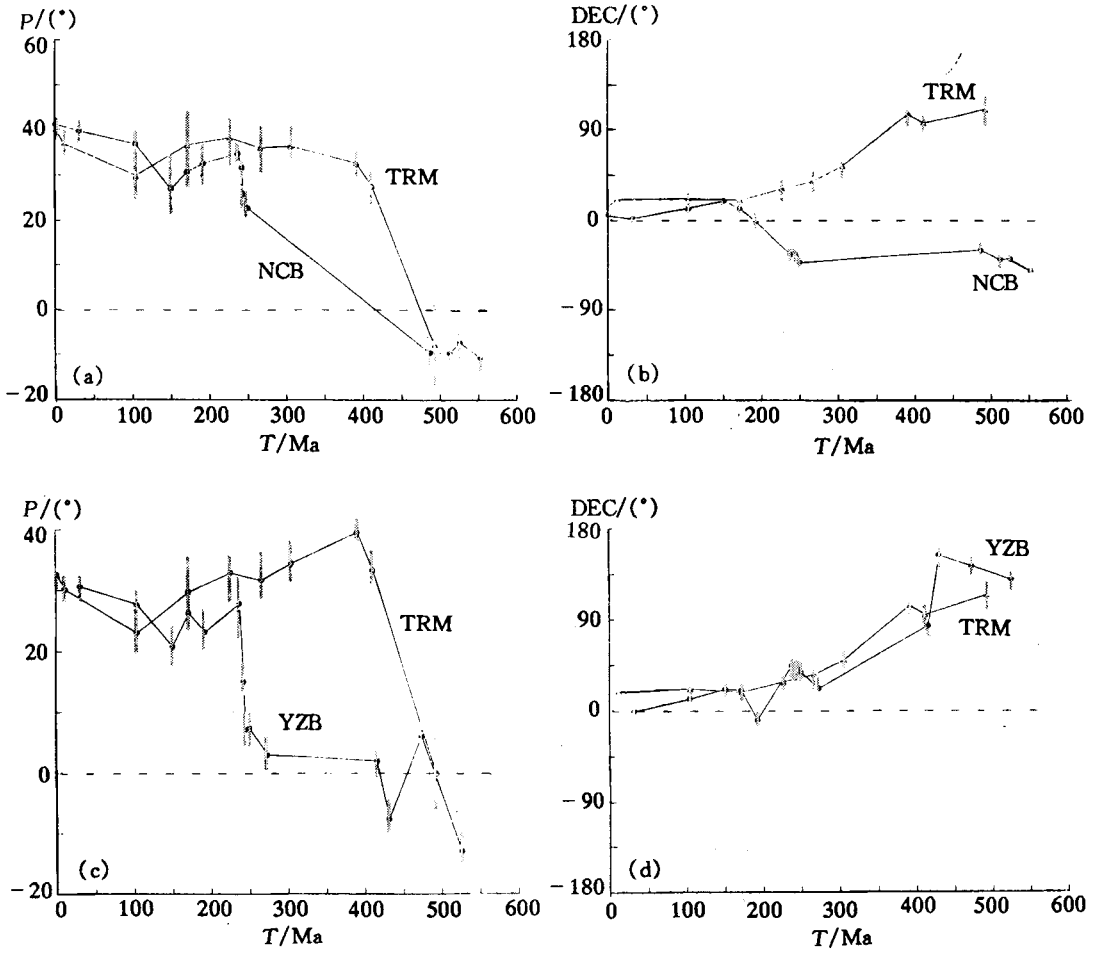


Fig. 5. Paleolatitudes and declinations deduced from the paleopoles of the TRM, NCB and YZB ((a)—(d)) at reference points at Jiayuguan (40°N/98°E) and Chengdu (31°N/104°E).

3.3 Tectonic evolution and paleocontinent reconstruction for the three blocks and their adjacent blocks

The existence of the Gondwanaland supercontinent is confirmed by paleomagnetic paleobiogeographic data. It is generally suggested that the Gondwanaland was formed in the Middle Cambrian[34,45], as a result of collision between proto-East and West Gondwana. However, some authors also argued that the Gondwanaland formed in the Late pre-Cambrian[34]. To clarify the tectonic history and conduct the paleocontinent reconstruction for the three major Chinese blocks, we introduced Siberia, Australia and India pole lists (table 5) which have been reexamined[34]. As shown in table 5, a large latitudinal placement of about 5 000 km, from low-latitude Southern Hemisphere to middle-latitude Northern Hemisphere during the Middle Cambrian to Silurian, was observed for Siberia. But no obvious rotation was revealed at the same time, indicating that Siberia might be independent of other blocks from the Middle Cambrian to Silurian and impossibly joined with Gondwanaland. The drift history was dominant by a rapid clockwise rotation (about 60°) for Siberia during the Silurian and Devonian. Siberia was basically stationary during the Devonian to Late Permian, while the NCB was obviously northward drifting in this period, indicat-

ing that the NCB and Siberia had not collided till the Late Permian. Siberia had northward drifted about 2000 km with a clockwise rotation by 24° during the Late Permian to Late Triassic, implying that Siberia had collided with Pangaea and TRM. While a paleolatitude discrepancy of about 20°—30° between NCB and Siberia existed in this period, indicating that the two blocks had not collided. During the Late Triassic to Late Jurassic, Siberia had southward drifted about 1500 km and collided with the North China and Mongolia Block[46]. This statement is now supported by the marine sediment discovered in Mongolia-Okhotsk tectonic unit, which was formed during the Triassic to Middle Jurassic[47]. It was this collision that caused the intracontinent orogeny discovered in the northern margin of the NCB in the Jurassic.

Australia was located near the equator in the Cambrian (table 5). Paleolatitudes of Australia and YZB were basically consistent in the Ordovician, a close biogeographic association of the YZB with Australia has been recognized for both the Cambrian and Devonian, suggesting that the YZB and Australia may have been parts of Gondwanaland in the Early and Middle Paleozoic[48]. Recently paleontological data from the TRM suggest that Cambrian and Ordovician Bradoriida fossils are very similar to those of the YZB, which would in turn suggest that the TRM was likewise part of Gondwanaland in the Early Paleozoic[9]. Paleomagnetic data show that they had been fragments from Gondwanaland during the Late-Middle Ordovician, and then Australia had southward drifted about 3 700 km from near the equator during the Ordovician to Middle Silurian, but the YZB about 1 400 km in this period. The drift history of Australia was characterized by both northward latitudinal displacement of 2 200 km and clockwise rotation by 62° during the Middle Silurian to Late Carboniferous, and a northward latitudinal displacement of about 2 200 km during the Middle Carboniferous to the Late Permian. Australia was northward drifting during the Late Permian to Early Jurassic, then it drifted slowly northward. India was basically standing at the low latitude of the Southern Hemisphere in the Early Paleozoic, and drifted southward about 3000 km and reached the middle latitude of the Southern Hemisphere during the Devonian to Late Permian. India has been northward drifting to the present-day position since the Cretaceous.

As shown in tables 4 and 5, the paleolatitudes and poles determined from Siberia and TRM had inclined to be consistent at 95% confidence level in Permian, indicating that two blocks had been sutured together in the pre-Permian. The collision produced a suture represented today by the Tianshan orogenic belt. Then TRM and Siberia drifted together northward with intracontinent rotation and wringing. During the Late Triassic, the Qiangtang Block collided with composite continent included the TRM and Siberia when it was drifting northward. The collision produced a suture represented today in the Kunlun Orogenic belt. The huge mosaic continent (combined by Qiangtang, TRM and Siberia) started southward drifting in the Middle Jurassic, and then collided with the NCB during the Late Jurassic[49]. Recently geologic data show that the eastward scissors-like trench in Mongolia-Okhotsk was closed in the Late Jurassic, supporting that Siberia and NCB/Mongolia were not sutured together till the Late Jurassic.

The lithospheric geodynamics indicates that the collision between India and Eurasia not only caused the uplift and crust shortening/thickening of the Qiangtang Block but also resulted in the clockwise rotation for Chinese continent[50]. For example the Ordos Basin rotated clockwise by 2—4 mm/a[50]. It should be noted that a clockwise rotation about 20° is noticeable for the three major blocks (NCB, YZB and TRM) with a quasi-constant velocity since the Late Jurassic (figs. 4, 5(b) and (d)). What did cause this rotation? The Lahsa Block and India subcontinent sub-

Table 5 Paleomagnetic data of India, Siberia and Australia[34]

India						Siberia						Australia					
Period	ϕp/(°E)	λp/(°N)	A_{95}/(°)	P/(°N)	D/(°)	Period	ϕp/(°E)	λp/(°N)	A_{95}/(°)	P/(°N)	D/(°)	Period	ϕp/(°E)	λp/(°N)	A_{95}/(°)	P/(°N)	D/(°)
N(3)	317	74		26.7	348.5	E(45)	177	78	11	41.2	15.7	E_1(45)	301	61	10	−12.7	168.8
E(50)	117	−25		22.8	161.3	K_1(121)	167	70	4	44.3	26.5	K(286)	318	56		−11.8	158.5
E(60)	285	43	10	−6.8	354.9	J_3(142)	150	70	9	49.9	24.8	K_1(122)	338	41	17	−7.6	138.7
K_2(95)	295	21	9	−27	342.2	J_2(173)	132	65	11	58.3	26.7	J_3(166)	349	48	13	−18.1	138.3
K_1(122)	299	12		−34.5	334.8	J_1(195)	110	63	21	65.9	13.4	J_1(185)	357	48	5	−22.3	134.8
J_3(165)	310	2		−38.8	317.2	T_3(207)	129	56	8	64.2	41.4	T_3(222)	350	32		−8	125.5
J_1(189)	310	10		−31.9	322.1	T_3(222)	138	55	9	59.7	46.9	T_1(240)	338	31		−0.2	132.1
T_3(200)	307	13		−30.5	326.7	T_2(238)	150	52	7	52.8	53.4	P_2(250)	321	36	21	4.3	146.4
T_2(225)	320	−9		−41.5	298.0	T_1(245)	156	52	4	49.2	53	P_1(270)	302	46		1.4	163.6
T_1(240)	305	−6	7	−48.2	317.3	P_2(253)	161	42	7	43.5	65.9	C_2(295)	322	62		−18	159.9
P_2(255)	283	4		−45.8	352.8	P_1(268)	159	38	9	43.5	71.8	C_1(335)	141	84		−44.3	185.7
C_2(340)	270	32		−17.6	7.1	C_3(288)	158	34	11	42.6	77.3	D_3(370)	203	62	16	−28.3	211
D_3(370)	243	22		−19.9	34.5	D(390)	151	28	8	45.1	88.5	S_2(420)	173	58	68	−40.6	222.4
D_1(385)	281	37		−13	357.5	S(420)	121	−4	19	41.2	148.8	O_1(480)	199	18	27	−3.4	249.3
S_2(420)	238	37		−4.7	31	O_2(460)	130	−22	4	21.2	148.2	Cam_3(530)	203	18	16	−0.6	246.7
O_3(440)	187	−14		−8.2	101.4	O_1(490)	132	−40	7	4.2	154.6						
O_1(450)	200	3		−7.2	79.9	Cam_2(530)	157	−44	12		141.3						
Cam_2(530)	199	6		−4.5	78.3												
Cam_1(560)	219	35	15	2.6	44.7												

N is the number of pole, reference point is placed at 40°N/98°E.

ducted successively northward and collided with Eurasia since the Jurassic, this geodynamic procedure conducted a tectonic stress force directed principally along SSW-NNE. Simultaneously, the collision between Pacific/Philippines plates and Eurasia made Chinese continent escape southeastward and spread to form an NNE structure line along the Pacific, which produced a force momentum drived clockwise rotation for Chinese continent. Recent observation using the GPS method in the Qiangtang Plateau provided more evidence for this clockwise rotation[1)].

Paleocontinents reconstruction during the Phanerozoic were presented based on paleomagnetic and other geologic data (fig. 6). However, paleolongitude of paleocontinents cannot be determined by paleomagnetic data, therefore longitude was arbitrary in figure 6.

4 Conclusion

The drift history of the three major blocks (NCB, YZB and TRM) and their adjacent blocks are concluded as the following based on paleomagnetic data (figs. 3—6). (i) Both the NCB and YZB were located at the mid-low latitude in the Southern Hemisphere during the early Paleozoic. The NCB was probably independent in terms of geodynamics, its drift history was dominant by latitudinal displacement and accompanying rotation in the Early Paleozoic. The YZB might be close to Gondwanaland in Cambrian, and separated from Gondwanaland during the Late-Middle Ordovician. (ii) TRM was part of Gondwanaland and close to the YZB and Australia in the Early Paleozoic. Paleomagnetic data show that they had been fragments from Gondwanaland during the Late-Middle Ordovician, and then the TRM drifted northward. The collision between the TRM and Siberia occurred during the Permian, but the composite North China and Mongolia Block did not collide with Siberia till the Late Jurassic. (iii) The paleomagnetic data revealed that the tectonic evolution of the NCB and YZB was characterized by northern latitudinal displacement and rotation around pivot at the Dabie Mt. from the Late Permian to Late Triassic. The NCB and YZB collided first in the eastern part where they were located at northern latitudes of about 6°—8°, and a triangular ocean remained between the NCB and YZB in the Late Permian. The collision propagated westward when the two blocks approached each other with a rotation of about 70° during the Late Permian and Middle Jurassic. It was such a collision and suturation procedure that produced intensive compression in the Dabie area between the NCB and YZB, where continent crust subducted into the upper mantle in the Late Permian, then the ultrahigh-pressure rocks cropped out in the collision zone between the NCB and YZB. Two blocks were drifting northward together, characterized by relative rotation and crust shortening. (iv) The NCB was rotated anticlockwise during the Late Permian to Early-middle Jurassic because the Mongolia Block continuously pushed on it, and the YZB was rotated clockwise at the same time. The NCB and YZB were sutured together in the Middle Jurassic, and then rotated anticlockwise at a slow velocity. The drift history of two blocks was characterized by northward latitudinal displacement and relative rotation accompanied by crust shortening, they reached the present-day position in the Late Jurassic. These statements indicate that the suturing zone between the NCB and YZB has diachronism, that is, not only the suturing zone was characterized by latitudinal displacement but also the crust shortening and relatively rotation took place. APWPs of the three major blocks are reflected by the systematic drift history of blocks both before collision and crust shortening/

1) Chen, Z.L., Crust motion of eastern Tibet determined by GPS, *Sci. in China* (in press).

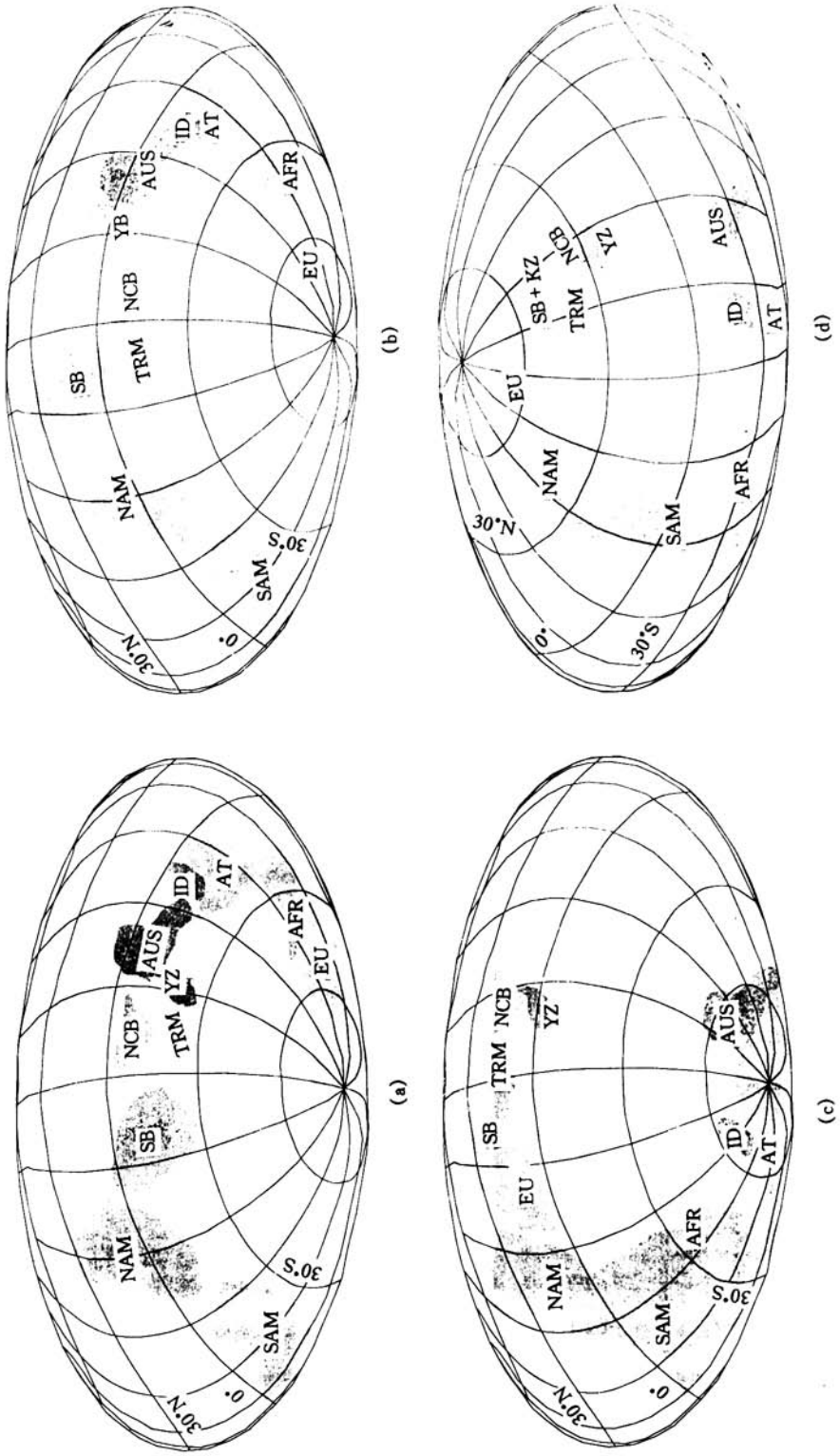


Fig. 6. Paleogeographic reconstruction showing the paleo-positions of the three major blocks. (a) Cambrian; (b) Ordovician; (c) Permian; (d) Jurassic. NCB, North China Block; YZB, Yangtze Block; TRM, Tarim Block; SB, Siberian Plate; KZ, Kazakstan Block; AUS, Australia Plate; SAM, South American Plate; NAM, North American Plate; AFR, African Plate; ID, Indian Plate; AT, Antarctic continent; EU, European paleocontinent.

relative rotation after suturing. Thus, the consistence of the APWPs determined from the three major blocks since the Late Jurassic indicates that they have become a dynamically unified block and no obvious crust shortening and relative rotation occurred between them. (v) The three major blocks (NCB, YZB and TRM) were rotated clockwise together about 20° relative to the present-day rotation axis since the Late Jurassic. The Lahsa Block and Indian subcontinent subducted successively to the north and collided with Eurasia since the Jurassic, this geodynamic procedure conducted a tectonic stress force directed principally along SSW-NNE. Simultaneously, collision between Pacific/Philippines plates and Eurasia made South China continent escape southeastward and spread, which form an NNE structure line along the Pacific, produced a force momentum driving China continent into rotating clockwise.

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